Cardiovascular Disease Prediction Using Statistical Learning and Neural Network Models

Alexandre Autran Léa Machefel

Abstract

We compare five methods for classifying the presence of cardiovascular disease in a public dataset of 70,000 patient records. The data contain one snapshot per patient and no follow-up, so the task concerns recorded disease status rather than future cardiovascular events. After cleaning and feature creation, 69,549 records remained. We compared Ridge and Lasso logistic regression, k-nearest neighbors, random forest, and a multilayer perceptron. Hyperparameters were selected by stratified 5-fold cross-validation on the training set, using the area under the ROC curve (AUC). Scaling was fitted within each training fold. The selected models were then evaluated on a separate test set. Random forest had the highest test AUC, 0.8052, followed closely by the neural network at 0.8026. Ridge and Lasso both reached 0.7956. Blood pressure, age, cholesterol, and body mass index carried much of the predictive information. The observed differences were small, while penalized logistic regression was easier to explain. These results describe performance within this dataset. External validation, a fuller calibration assessment, and evaluation in a defined clinical setting are needed before any practical use.

Keywords: cardiovascular disease classification; statistical learning; logistic regression; random forest; neural network; interpretability.

1 Introduction

Cardiovascular diseases are the leading cause of death worldwide [2]. Models based on routine measurements may help researchers study how these measurements relate to recorded disease. Their usefulness depends on the quality of the data, the target being predicted, and the setting in which they are tested.

Here, the target is $Y = \texttt{cardio} \in \{0, 1\}$, and the models estimate $\widehat{p}(X)$, an estimate of $P(Y = 1 \mid X)$ in the study data. The predictors include age, body measurements, blood pressure, cholesterol, glucose, and a few reported behaviors. Each row is a single patient record. There is no follow-up period or time to an event. The models therefore classify the dataset's disease label; they do not estimate a person's future cardiovascular risk.

This project has two aims. The first is to provide a reproducible analysis, from data cleaning to final evaluation. The second is to compare simple logistic models with more flexible methods: k-nearest neighbors (KNN), random forest, and a multilayer perceptron (MLP). We consider predictive performance alongside the difficulty of explaining the models. The accompanying Python scripts contain the implementation [1].

2 Dataset and Exploratory Analysis

2.1 Dataset structure

The original Kaggle dataset contains 70,000 records and no explicit missing values [3]. Its columns are `age` in days, `height` in centimeters, `weight` in kilograms, `gender`, systolic blood pressure (`ap_hi`), diastolic blood pressure (`ap_lo`), `cholesterol`, `gluc`, `smoke`, `alco`, `active`, and the binary target `cardio`. The identifier `id`, when present, is excluded from the predictors. Cholesterol and glucose use ordered categories; smoking, alcohol use, and physical activity use binary indicators.

The absence of missing entries does not guarantee valid measurements. Some blood-pressure values were negative, reversed, or far outside the expected range. We applied the cleaning rules described below, converted age to years, and calculated body mass index (BMI). The final dataset contained 69,549 records: 451 records, or 0.64%, were removed. We retained BMI and height and removed raw weight from the final predictor set.

2.2 Variables associated with the disease label

Records with `cardio`=1 tended to have higher age, blood pressure, and BMI. Higher cholesterol and glucose categories were also more common in this group. The distributions overlapped, however, so none of these variables could identify disease status on its own.

The largest Pearson correlations with the binary target were approximately 0.433 for systolic pressure, 0.347 for diastolic pressure, 0.239 for age, 0.221 for cholesterol, and 0.195 for BMI. These describe separate, marginal associations. For ordinal variables, their size also depends on the numerical coding. They do not establish causal effects.

Figure 1 summarizes these relationships. Glucose showed a smaller positive association with the target. Gender and the three behavioral indicators had weaker marginal associations. Their limited detail is one possible explanation: a binary smoking indicator, for example, does not describe duration or amount. A weak association in this dataset should not be read as evidence that a behavior is unimportant to cardiovascular health.

2.3 Atypical measurements

The raw blood-pressure columns contained zeros, negative values, very small values, and extreme values. The cleaning procedure treated some of these entries as possible sign or scale errors. Corrections and exclusions were based on explicit numerical rules, rather than confirmation from the original patient records. Corrected values

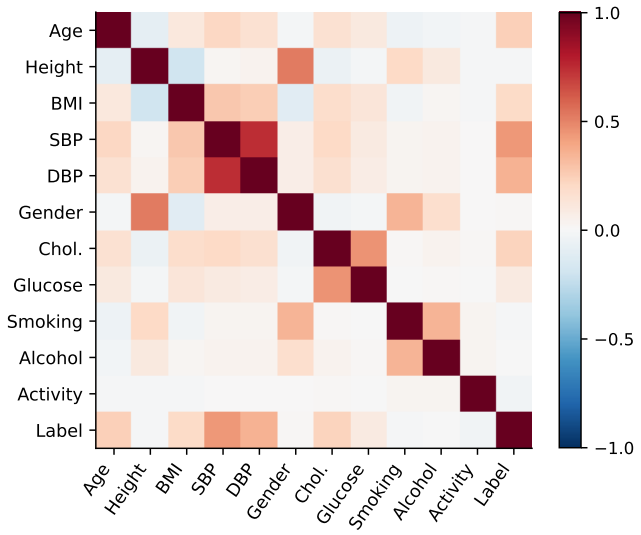


Figure 1: Pearson correlations after cleaning and feature creation. SBP and DBP denote systolic and diastolic blood pressure. The label is `cardio`.

may therefore still contain error.

After cleaning, systolic pressure ranged from 60 to 240 mmHg, with a mean of 127.04 mmHg. Diastolic pressure ranged from 40 to 150 mmHg, with a mean of 81.57 mmHg. Mean age was 53.30 years, and mean BMI was 27.42 kg/m^2. The BMI distribution remained right-skewed.

2.4 Relationships between predictors

Systolic and diastolic pressure were positively correlated. Weight and BMI were also strongly related because BMI is calculated from weight and height:

$$\mathrm{BMI} = \frac{\text{weight (kg)}}{[\text{height (cm)}/100]^2}.$$

We kept BMI and height and dropped raw weight. This reduces the number of inputs, although BMI and height together still contain enough information to reconstruct weight. The choice is therefore a change in representation rather than the removal of all shared information.

2.5 Class balance and evaluation measures

The cleaned dataset contained 34,823 negative labels and 34,726 positive labels, corresponding to 50.07% and 49.93%. An always-negative classifier would have an accuracy of 0.5007. These proportions describe the study sample and should not be interpreted as population disease prevalence.

We report accuracy, precision, recall, F1-score, AUC, and a confusion matrix. For the positive class,

$$\mathrm{precision} = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FP}}, \qquad \mathrm{recall} = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FN}},$$

$$\mathrm{F1} = \frac{2\,\mathrm{TP}}{2\,\mathrm{TP}+\mathrm{FP}+\mathrm{FN}}.$$

A false negative is a record with `cardio=1` that is classified as negative; a false positive is the reverse. In a real screening setting, these errors could have different consequences. Those consequences were not measured here.

3 Preprocessing Pipeline

3.1 Cleaning and feature creation

Columns imported as text, including weight and blood pressure, were converted to numeric values. All negative blood-pressure entries were replaced by their absolute values. Values of at least 1,000 were divided by ten. Systolic values from 11 to 25 and diastolic values from 4 to 15 were then multiplied by ten. A reversed pair was swapped when the resulting systolic value was 60–250 and the resulting diastolic value was 40–150 mmHg. The scripts provide the ordered rules [1].

The retained ranges were 18–100 years for age, 60–250 mmHg for systolic pressure, 40–150 mmHg for diastolic pressure, 130–210 cm for height, 40–200 kg for weight, and 15–50 kg/m^2 for BMI. Systolic pressure also had to be at least as high as diastolic pressure, and the target had to be 0 or 1. These are study-specific filters. They are not universal boundaries between possible and impossible adult measurements: some excluded values could represent real patients. Replacing signs or rescaling values also assumes a particular recording error that could not be checked against source records. The reported results therefore apply to the retained sample. Age was divided by 365.25, and BMI was calculated before raw weight was removed.

3.2 Training and test sets

We used a stratified 80%/20% split, giving 55,639 training records and 13,910 test records. The positive-label proportion rounded to 0.4993 in both sets. Stratification preserved the near balance of the original sample.

Hyperparameters within each model family were selected on the training set by stratified 5-fold cross-validation. The main comparison used the default classification threshold of 0.5. The five fitted candidates were evaluated on the test set. The scripts then ranked them by test AUC and used the first model for the confusion-matrix and threshold plots. The test set was therefore used to identify the reported winner, although it was not used to tune the within-family hyperparameters. Its score is not an independent estimate of the performance of that model-selection procedure. An independent test set would be needed for that purpose.

The split, cross-validation shuffling, and stochastic estimators used `random_state=42`. The code and public dataset are available through the cited sources [1,3]. The analysis used data accessed on 16 October 2026, with NumPy 2.5.4, pandas 3.0.6, and scikit-learn 1.9.1. For compatibility with pandas 3, the blood-pressure columns were converted to floating-point values before division.

3.3 Scaling and coding

Age, height, BMI, and the two blood-pressure variables were standardized as

$$z = \frac{x - \mu_{\text{fit}}}{\sigma_{\text{fit}}},$$

where the mean and standard deviation came only from the observations used to fit that step. During cross-validation, the scaler was fitted on each training fold and applied unchanged to its validation fold. For the final evaluation, it was fitted on the full training set and applied to the test set.

Scaling matters for KNN, penalized logistic regression, and the MLP because these methods depend on feature magnitudes. The same pipeline was also applied to random forest for consistency, although tree-based splits do not require standardization. The remaining predictors kept their numerical coding. Cholesterol and glucose retained their ordered categories. In logistic regression, this assumes the same change in log-odds for each one-level increase.

3.4 Preventing information leakage

Data-dependent transformations were fitted inside the cross-validation pipeline, using training observations only. Validation and test observations were transformed without refitting the scaler. This prevents their values from influencing the scaling used for their predictions.

The numerical cleaning rules were applied before splitting and did not use the target to modify predictors. The separate label check only retained valid binary targets. These checks differ from estimating a transformation from the data. However, changes to cleaning rules or feature choices after inspecting test results would still require a fresh test set. The supplied scripts do not document when those choices were made.

4 Models

We compared two penalized logistic models, KNN, random forest, and an MLP [5, 9]. Hyperparameter selection used mean cross-validation AUC. AUC measures how well a model ranks positive records above negative records across score thresholds [12]; it does not measure calibration or clinical usefulness.

4.1 Ridge logistic regression

Logistic regression models the log-odds of the positive label as a linear function of the inputs. Ridge adds an L_2 penalty:

$$\min_{\beta_0,\beta} \left[-\ell(\beta_0, \beta) + \lambda \sum_{j=1}^{p} \beta_j^2 \right],$$

where ℓ is the log-likelihood and λ controls the penalty. This shrinks coefficients and can help when inputs are correlated [6, 9]. The implementation uses C, an inverse regularization parameter; its exact relation to λ depends on the loss convention.

We tested $C \in \{10^{-3}, 10^{-2}, 10^{-1}, 1, 10, 100\}$. Smaller C gives a stronger penalty. Cross-validation selected $C = 0.1$, with mean AUC 0.7929.

4.2 Lasso logistic regression

Lasso uses an L_1 penalty. With the `liblinear` solver and its default intercept scaling, the fitted objective also penalizes the intercept:

$$\min_{\beta_0,\beta} \left[-\ell(\beta_0, \beta) + \lambda \sum_{j=0}^{p} |\beta_j| \right].$$

The penalty can set coefficients to zero [7, 17]. Cross-validation selected $C = 0.1$, with mean AUC 0.7929. All 11 predictor coefficients remained nonzero, so the fitted Lasso model did not remove any predictors. Its results were close to Ridge.

4.3 k-nearest neighbors

KNN uses nearby training records to predict a new record [8]. With uniform weights, its positive-class score is

$$\hat{p}(x) = \frac{1}{k} \sum_{i \in \mathcal{N}_k(x)} y_i.$$

We tested $k \in \{20, 50, 100\}$, using Euclidean distance and uniform weights throughout. The selected model used $k = 100$, with mean cross-validation AUC 0.7966. Small neighborhoods can follow noise; large neighborhoods produce smoother predictions and may miss local patterns.

4.4 Random forest

A random forest combines many decision trees trained with random sampling of records and candidate predictors [4]. The predicted probability averages the trees' probabilities. Trees can represent nonlinear relationships and interactions without specifying them in advance.

The search varied the number of trees, maximum depth, and minimum leaf size. The selected forest used 300 trees, maximum depth 12, and `min_samples_leaf`=10. Its mean cross-validation AUC was 0.8021.

4.5 Multilayer perceptron

The MLP uses fully connected hidden layers with ReLU activations and a sigmoid output for binary classification. Its weights are fitted by backpropagation [10, 11]. We tested hidden-layer sizes (50), (100), and $(50, 25)$. The selected network used $(50, 25)$, an L_2 regularization parameter `alpha`=0.001, and an initial learning rate of 10^{-3}. Mean cross-validation AUC was 0.8017.

The data-fitting part of the loss is binary cross-entropy:

$$\mathcal{L}_{\text{CE}}(\theta) = -\frac{1}{n} \sum_{i=1}^{n} \big[y_i \log \hat{p}_\theta(x_i) + (1 - y_i) \log(1 - \hat{p}_\theta(x_i)) \big].$$

The fitted objective also includes an L_2 penalty on the network weights, controlled by `alpha`. Early stopping uses an internal validation subset of the current training data.

Table 1: Performance of the selected models. CV AUC is the mean and standard deviation across five training folds. The standard deviation is not a confidence interval.

Model	Acc.	Prec.	Recall	F1	Test AUC	CV AUC
Random forest	0.7393	0.7638	0.6917	0.7260	0.8052	0.8021 ± 0.0043
MLP $(50, 25)$	0.7358	0.7663	0.6775	0.7191	0.8026	0.8017 ± 0.0047
KNN $(k = 100)$	0.7347	0.7654	0.6759	0.7178	0.7968	0.7966 ± 0.0045
Ridge	0.7322	0.7615	0.6750	0.7157	0.7956	0.7929 ± 0.0043
Lasso	0.7323	0.7617	0.6749	0.7157	0.7956	0.7929 ± 0.0043

In scikit-learn's `MLPClassifier`, the stopping score is validation accuracy, even when the outer hyperparameter search uses AUC [16].

5 Results

Table 1 reports test performance after cross-validation selected one candidate per model family. The scripts obtain classes through each estimator's `predict` method, which uses its default class decision rule. This generally corresponds to a probability threshold near 0.5, with implementation-specific handling of exact ties. AUC is calculated from the scores and does not require a single threshold.

Random forest had the highest test AUC, followed closely by the MLP. The full AUC range was 0.7956–0.8052, an absolute difference of 0.0096. Cross-validation and test AUC values were close, but this agreement does not prove that the models are free from overfitting. No confidence intervals or paired comparison tests are reported, so the small observed differences should not be treated as established differences in population performance.

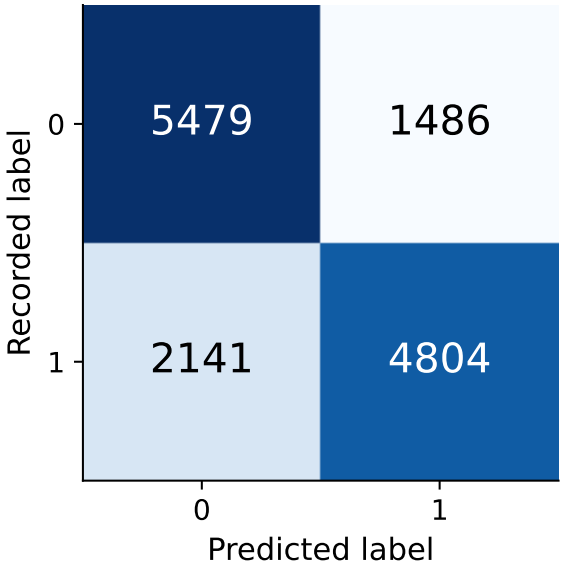


Figure 2: Random-forest confusion matrix on the test set at threshold 0.5. Errors refer to the recorded disease label.

5.1 Probability quality and calibration

Calibration concerns agreement between predicted probabilities and observed positive-label frequencies [14]. We also report the Brier score,

$$\mathrm{BS} = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{p}_i)^2,$$

which was 0.1787 for random forest and 0.1851 for Ridge. Lower values indicate better overall probability predictions on the same data. The Brier score reflects both calibration and discrimination, so a lower value alone does not establish better calibration [18]. Reliability curves, calibration intercepts and slopes, and independent validation would give a fuller assessment. These scores are not yet validated estimates of an individual's future medical risk.

5.2 Random-forest errors and threshold choice

At threshold 0.5, the random-forest confusion matrix was

$$\begin{pmatrix} 5479 & 1486 \\ 2141 & 4804 \end{pmatrix},$$

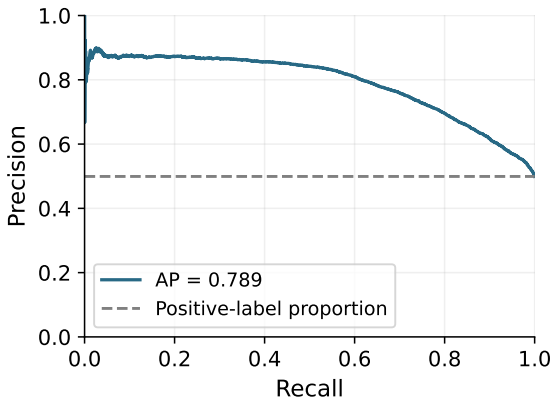
with true classes in rows and predicted classes in columns, both ordered as 0 and 1. There were 5,479 true negatives, 1,486 false positives, 2,141 false negatives, and 4,804 true positives. Recall was $4804/(4804 + 2141) \simeq 0.6917$.

On the same test set, lowering the threshold to 0.40 gave recall 0.7826 and precision 0.7090. More positive labels were detected, at the cost of more false positives. The script evaluates thresholds from 0.30 to 0.60 on this test set, so these results are an exploratory sensitivity analysis. They do not validate a clinical threshold. A threshold intended for practice should be selected using training or validation data, with explicit attention to the setting and the consequences of errors, then assessed on independent data.

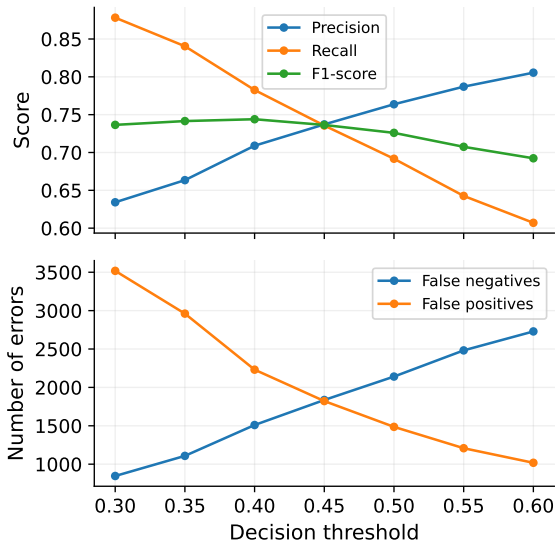
The precision–recall curve and threshold plots in Figure 3 make this trade-off visible [13]. Precision also depends on the positive-label proportion, so it may change substantially in a population with a different prevalence.

5.3 Stability and uncertainty

Cross-validation and test scores followed the same broad pattern. A single split and fixed seed do not measure stability across samples. The fold standard deviations are not confidence intervals for test AUC. Repeated splits or paired resampling would help assess uncertainty in the model comparison.



(a) Precision–recall curve.



(b) Measures across thresholds.

Figure 3: Threshold analysis for random forest. A lower threshold detects more positive labels and produces more false positives.

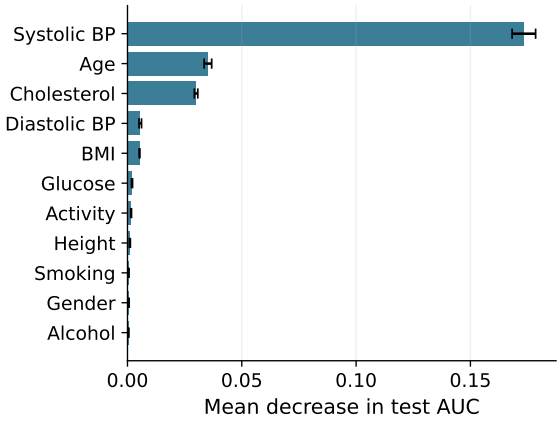


Figure 4: Random-forest permutation importance, measured as the mean decrease in AUC after shuffling each predictor. Error bars show the standard deviation over five permutations. Importance describes the fitted model's use of a variable, not its causal or clinical importance.

5.4 Interpretability

Ridge and Lasso provide one global equation. A coefficient's sign describes the direction of the fitted association while holding the other model inputs fixed. Coefficients for standardized continuous variables refer to one standard-deviation changes. Binary and ordinal coefficients use different units, so their magnitudes should not all be compared directly. None of these coefficients estimates the effect of an intervention.

For random forest, permutation importance compares the original AUC with the AUC after shuffling one predictor:

$$\Delta\text{AUC}_j = \text{AUC} - \text{AUC}_{\pi(j)}.$$

Systolic pressure, age, cholesterol, diastolic pressure, and BMI had the largest values (Figure 4). The scripts repeated each permutation five times on the test set. This gives a direct view of the fitted model's reliance on each input [4, 15]. Correlated predictors can share information, which can make either one appear less important. Shuffling can also create combinations that are uncommon in real records. The scores therefore need careful interpretation.

The forest's global importance scores do not fully explain an individual prediction. The MLP is harder to summarize because its hidden units have no direct clinical meaning. KNN can be described through nearby records, although the selected neighbors depend on the scaling and distance definition.

6 Discussion

Blood pressure, age, cholesterol, and BMI showed strong associations with the target and contributed useful predictive information. These patterns are broadly consistent with established cardiovascular risk factors [2]. However, the study does not show when disease developed or whether measurements preceded diagnosis. Existing disease or treatment may affect the observed measurements and behaviors. The results therefore describe classification within this dataset, rather than the effects of changing a person's risk factors.

6.1 Comparison of methods

Penalized logistic regression already captured much of the available signal. Ridge and Lasso had almost identical results and retained all 11 predictors. Their main advantage is a compact model that can be inspected through its coefficients. This clarity does not make the model automatically correct or clinically useful.

Random forest achieved the highest AUC in this analysis. Its ability to model nonlinear relationships and interac-

tions may explain the modest gain, although we did not isolate which features of the model produced it. The MLP performed almost as well. Neither result demonstrates that these model families will be superior on a different sample or population.

6.2 Dataset limitations

The public data contain one simplified record per patient. They lack follow-up, diagnosis dates, disease severity, medication details, family history, and detailed past exposures. The available documentation does not provide enough clinical detail to establish the label's diagnostic standard or the timing of measurements relative to disease. The label itself should therefore be treated as a dataset outcome, rather than a fully characterized clinical endpoint.

The near 50% positive-label proportion does not establish representativeness. Cleaning may also change the population represented by the retained records. We did not assess performance across patient subgroups, test on an external dataset, or quantify the clinical consequences of using the predictions. These gaps limit any claim about screening or individual decision support.

7 Conclusion

The five models classified the recorded cardiovascular disease label with similar performance. Random forest had the highest observed test AUC, 0.8052, while Ridge and Lasso reached 0.7956. Blood pressure, age, cholesterol, and BMI provided much of the predictive information. The small performance differences leave room for a simple logistic model when a clear explanation is a priority. Random forest remains an alternative to examine in independent validation.

These results describe a cleaned public dataset. Independent validation is especially important because test AUC was used to rank the model families and the same test data were used to explore thresholds. Uncertainty estimates, a fuller calibration assessment, and threshold selection for a defined setting are also needed before practical use. The code supports checking and extending the analysis [1].

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